Healthcare Insurance Claims Fraud Detection

System Architecture & Implementation Plan

Version 1.0 | Generated 2026-10-29 | Random seed 20261029

Purpose. Define a local, reproducible POC that generates fictional healthcare claim data, identifies suspicious patterns using a hybrid detection approach, and presents evidence to a human reviewer.

1. Architecture overview

Layer	Components	Outputs
Synthetic data	Python, NumPy, Pandas; members, providers, procedures, claims, line items	Versioned CSV tables and separate synthetic labels
Data validation	Schema checks, null checks, referential integrity, date and amount validation	Validated claim dataset and quality summary
Feature engineering	Duplicate keys, claim velocity, billed/reference ratios, provider aggregates	Model-ready feature matrix; no label columns
Detection	Deterministic rules, provider peer groups, Isolation Forest	Rule flags, anomaly score, reason codes
Risk prioritization	Configurable weighted score and review bands	Ranked investigation queue
Streamlit app	Dashboard, claim detail, provider analytics, review outcomes	Interactive local POC
Evaluation & governance	Synthetic-label evaluation, false-positive review, threshold analysis	Metrics, charts, audit-friendly explanations

2. Data flow

CSV source tables	→	Validation + feature engineering	→	Rules + anomaly model
Separate labels	→	Evaluation only after scoring	→	Metrics and threshold review

3. Technology choices

Technology	Use in POC
Python / Pandas / NumPy	Data generation, validation, transformations and aggregation
scikit-learn Isolation Forest	Unsupervised anomaly ranking on numeric/encoded features
Streamlit + Plotly	Interactive dashboard, filters, claim-level drill-down and charts
CSV	Transparent local persistence for first iteration
ReportLab + Matplotlib	PDF documentation and static supporting charts

4. Key design controls

Label leakage prevention

synthetic_labels.csv is excluded from feature engineering, rule execution, model fitting and claim scoring. It may be joined only after scoring for evaluation.

Human-in-the-loop

A risk flag is a review signal, not a determination of fraud, wrongdoing, payment eligibility or provider intent.

Reproducibility

Fixed NumPy random seed: 20261029. Identical versions and parameters should produce repeatable data.

Synthetic-only scope

All member, provider, claim, diagnosis and procedure records are fictional. Amounts and rules are illustrative, not official reimbursement guidance.

Threshold governance

Risk bands and weights are initial configurable POC assumptions. They require validation before any real-world use.

Privacy and security

No real protected health information is required. Do not replace synthetic data with real patient data without an approved privacy/security design.

5. Proposed risk score

Signal	Initial weight	Rationale
Duplicate indicators	25%	Repeated claim identity and service characteristics
Unusual billed amount	20%	Amount deviation from procedure or peer benchmark
Upcoding-like indicators	15%	Illustrative procedure/units mismatch signals
Provider peer-group outlier	15%	Provider-level aggregate deviation
Claim frequency	10%	Repeated services or unusually high units
Isolation Forest	15%	Multivariate anomaly signal

Initial display bands: Low 0–39, Medium 40–69, High 70–100. These bands are illustrative, are not fraud probabilities, and should be tuned against validation results.